MIL-V-23289B

20 August 1976
SUPERSEDING
MIL-V-23289A
5 September 1968

MILITARY SPECIFICATION

VALVES, SOLENOID, REFRIGERANT-12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solenoid valves for use with refrigerant-12.

1.2 Classification. Valves shall be of the following types, sizes, styles, and electrical ratings, as specified (see 6.2):

Type I - For liquid refrigerant
  Size 1 - 3.0 tons of refrigeration (TR) (10548 watts (W)) minimum capacity
  Size 2 - 7.5 TR (2637 W) minimum capacity

Type II - For refrigerant gas
  Size 3 - 2.5 cubic feet per minute (CFM) (.0127 metre per second (M^3/s)) minimum capacity
  Size 4 - 5.0 CFM (.0254 M^3/s) minimum capacity

Style 1 - Without manual opening stem
Style 2 - With manual opening stem

208 volts - 60 Hertz
120 volts - 60 Hertz
24 volts - DC
12 volts - DC

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command (GL), Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein), Packaging Of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MS 17841 - Valves, Solenoid, Refrigerant-12.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

Underwriters' Laboratories, Inc. (UL)

UL 429 - Electrically Operated Valves

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville, LI, NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050; 333 Pfingsten Rd., Northbrook, IL 60062; or 2602 Tampa East Blvd., Tampa, FL 33619.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Standard product. The valves delivered under this specification shall be the manufacturer's commercially standard product, except for any changes necessary to comply with the specification requirements.

3.2 Safety standard. The valves shall conform to the applicable requirements of UL 429.

3.2.1 UL certification. Prior to approval of the first shipment, the contractor shall submit to the contracting officer or his authorized representative, satisfactory evidence that the valves he proposes to supply under this specification meet the applicable requirements of UL 429. Acceptable evidence of meeting the applicable requirements of this standard shall be the UL label, a listing in the UL Electrical Materials Construction List, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the refrigerant containing components offered have been tested and conform to the applicable requirements of UL 429.

3.3 Materials. Materials not definitely specified shall be of the quality normally used for the purpose by the contractor.

3.4 Design and construction. The design and construction of the valves shall be in accordance with the applicable requirements of MS 17841.

3.5 Marking. In addition to the marking required by UL 429, the valves shall be marked with the Maximum Operating Pressure Differential and the direction of flow of the refrigerant.

3.6 Finish. The finish on the valves shall be the manufacturer's standard finish.

* 3.7 Workmanship. All parts of the finished valves shall be clean and free from pits, sprues, scale, flux or extraneous material. External surfaces shall be free from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with provisions set forth in MIL-STD-105.

* 4.2.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification.

4.2.2 End item inspection. The inspection lot shall consist of all valves of one type, size, style and electrical rating offered for inspection at one time. The sample unit for this examination shall be one valve.

* 4.2.2.1 Visual examination. The valve shall be examined for defects in table I. The inspection level shall be II, and the acceptable quality level (AQL) shall be 2.5 for major and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Visual classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished in accordance with manufacturer's commercial practice.		X
Design, construction and workmanship	Contains flux or extraneous material, not clean, has pits, sprues or scale.		X
	Contains burrs, slivers, or sharp edges.		X
Marking for identification	Missing, incomplete, illegible, not marked in prescribed manner.	X	

4.2.2.2 Dimensional examination. Examination shall be made of the valves to determine compliance with dimensions specified on MS 17841. Any dimension not within the specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

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4.2.3 Examination of preparation for delivery. An examination will be made to determine that packaging, packing, and marking are in compliance with MIL-V-3.

* 4.3 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

5. PACKAGING

* 5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-V-3.

6. NOTES

6.1 Intended use. The solenoid valves covered by this specification are for use as refrigeration system controls.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, size, electrical rating, and style required (see 1.2).

6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 82

Preparing activity:

Army - GL

Project No. 4130-0232

Review activities:

Army - MD
DSA - GS

User activities:

Army - MI, CE
Navy - SH

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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